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Rapid Evidence Synthesis: Remote Spirometry

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Rapid Evidence Synthesis:

Rapid Evidence Syntheses (RES) are produced by the National Institute for Health and Care Research (NIHR) Applied Research Collaboration Greater Manchester (ARC-GM). The methods used are based on a framework set out in Norman et al. 2022 and previously registered on the Open Science Framework (OSF).[a,b]

RES use evidence synthesis approaches and draw on the GRADE Evidence to Decision framework[c] to provide rapid assessments of the existing evidence and its relevance to specific decision problems. In the first instance they focus on evidence from guidance and existing evidence syntheses. They are undertaken in a real-time context of decision-making around adoption of innovative health technologies and are designed to provide a "good-enough" answer to inform decision problems in a short timescale. RES methods are flexible and adaptive. They have evolved in response to user feedback and differ depending on the nature of the assessment undertaken.

RES are not intended to serve as a substitute for a systematic review or rapid review of evidence.

We welcome feedback and are particularly interested to hear how you have used this Rapid Evidence Synthesis.

Please send any queries or comments to:

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This work was undertaken by the National Institute for Health Research (NIHR) Applied Research Collaboration Greater Manchester (ARC-GM). The views expressed are those of the author and not necessarily those of the NIHR or the Department of Health and Social Care.

[a] Norman, G. Rapid evidence synthesis to support health system decision making. *OSF registration.* 2020 [cited 2023]; Available from: osf.io/hsxk5

[b] Norman, G., et al., Rapid Evidence Synthesis To Enable Innovation And Adoption in Health and Social Care. *Systematic Reviews,* 2022. 11: p. 250. https://doi.org/10.1186/s13643-022-02106-z

[c] Alonso-Coello, P., et al., GRADE Evidence to Decision (EtD) frameworks: a systematic and transparent approach to making well informed healthcare choices. 1: Introduction. *BMJ,* 2016. 353: p. i2016.

1. Summary

There is **indirectly relevant evidence** for the use of remote supported spirometry at home in people with asthma or COPD; **an ongoing study which will supply directly relevant evidence has not yet reported.**

There is **considerable indirectly relevant evidence which reports some relevant data** for the comparability of measurement properties and some for patient experience.

There is **indirectly relevant evidence which is of very low to moderate certainty** for the impact on patient health outcomes. The evidence is impacted by indirectness, some inconsistency and incompleteness of results and methods which are not always well reported.

1.1 Measurement performance

There is evidence from three separate sources for measurement performance of supported home spirometry. These are (i) in clinic validation of devices under consideration in relevant populations; (ii) unsupported home spirometry with other/unspecified devices in relevant populations; (iii) a small in-clinic simulation of supported home spirometry with devices under consideration in an indirectly relevant population. **A small, directly relevant ongoing study was identified but has not yet posted results.** It should be noted that much of this data relates to people who are either diagnosed with asthma/COPD/another lung condition or have a sufficient need to be referred to a hospital lab service.

1.1.1 In-clinic validation

In-clinic validation has evidence from two studies of 200 people each who had asthma, COPD, other/unstated lung disease or were healthy volunteers; each looking at a different version of the NuvoAir device. These showed some systematic differences between the new devices and standard lab-based spirometry. Different measures were used in the two studies, but both concluded that agreement was acceptable; details of missed diagnoses were reported as approximately 10% for one study and the other reported “the majority of measurements were well within the limits of agreement”.

These studies were undertaken in hospital clinics rather than using the remote/home testing conditions in which the devices would be used in practice, but they are otherwise likely to be relevant. Some methods and some useful data (exact numbers within limits of agreement and threshold sensitivity and specificity) are not reported in each case.

1.1.2 Unsupported spirometry

Unsupported spirometry undertaken daily at home using other devices was used in three medium or large RCTs of other interventions in people with asthma or COPD. These studies also found that measurements (primarily FEV_1, FVC, FEV_1/FVC ratio) differed from laboratory measurements but differed as to whether the differences were considered significant. One study reported 6% of participants had measurements outside the limits of agreement and recommended against home spirometry with the device used; other studies considered differences to be present but consistent and minimal. Differences in devices used are likely to be responsible for the different findings in these large studies using repeated measures embedded within randomised trials.

Although the population and setting are relevant the devices and absence of support make this indirectly relevant.

1.1.3 Supported spirometry at home

There is indirectly relevant evidence from an in-clinic simulation with the device being considered in 40 people with motor neurone disease or similar conditions. The ability to detect lung capacity reduction was threshold-dependent and the measure reported (FVC) was overestimated at all levels but more so at higher levels of lung capacity.

This is **a small study using a simulation of home spirometry in a very different clinical population** to people with known or suspected asthma or COPD.

1.2 Impact on outcomes

The only evidence we identified is indirectly relevant as it relates to the impact of ongoing (continuous) monitoring involving home spirometry. Evidence from a recent Cochrane systematic review of monitoring in people with diagnosed COPD found moderate certainty evidence that patients who have asynchronous remote monitoring plus usual care or as part of a multi-component intervention have a longer time before hospital readmission than those in the control group.

All evidence for other outcomes and other comparisons was low or very low certainty.

1.3 Patient experience

There are some data from a small in-clinic simulation of supported home spirometry in people with motor neurone or related disease; this indicated positives for patients/carers and therapists as well as qualitative data showing identified issues or concerns, particularly from the therapists. A small, directly relevant ongoing study was identified but has not yet posted results.

2. Methods

2.1 Description of the Intervention

Spirometry is a test used to assess lung function. It is used as part of the diagnostic process for long term respiratory conditions including asthma and COPD, and also to monitor these conditions. It involves blowing into a device in specific ways to determine the amount of air which can be breathed out in a single breath. It is usually carried out in a clinic or general practice setting with the support of a doctor, nurse or other clinician.

Remote spirometry in the approach being considered for implementation in Greater Manchester involves home delivery of the spirometry hand-held device together with a smartphone with a pre-installed app to the patient's home. The patient is guided through the spirometry tests over a video call with a physiologist. Once the test is completed data are accessible immediately to the physiologist, and subsequently to the patient's doctors. Devices are returned by courier or mail after testing is completed.

2.2 Key Questions

1) What is the measurement performance of remote spirometry compared to spirometry performed in a clinical or laboratory setting? Lab-based spirometry is the gold standard but an appropriate comparator for this RES is spirometry performed in clinic.

a) For patients with suspected or diagnosed asthma?
b) For patients with suspected or diagnosed COPD?

2) What is the impact of remote spirometry on patient health outcomes compared to usual care or to spirometry performed in a clinical setting.

a) For patients with suspected or diagnosed asthma?
b) For patients with suspected or diagnosed COPD?

Appropriate outcomes for patients with respiratory disease usually include mortality (overall and condition-specific), unscheduled use of health care including emergency department attendance and hospital admissions, in addition to symptoms (e.g. breathlessness, night waking, limitation of activity) and quality of life. Medication and other management changes were also considered.

3) What is patients' experience of using remote spirometry and what are the barriers and facilitators to use.

a) For patients with suspected or diagnosed asthma?
b) For patients with suspected or diagnosed COPD?

Where there is no or limited evidence relating to remote spirometry we considered evidence relating to other remote diagnostic testing or health assessment for patients with respiratory conditions; home testing using digital technology for other conditions were also considered, where necessary.

2.3 Search

We searched Medline (Ovid) and the Cochrane Library (including the Cochrane Database of Systematic Reviews and the Cochrane Register of Controlled Trials (CENTRAL)) in June 2023.

We used a Medline search strategy based on the facet of spirometry and other respiratory testing terms and combined this using AND with a set of terms for the facet of remote consultation. We used additional terms for home-based provision but added a systematic reviews filter for this element of the search. The Cochrane search strategy used the two main facets described above.

We will check references of identified studies and where appropriate use forward citation searching. We will also check the websites of the innovation sponsor and key organisations such as NICE and NHS England. We will make use of material supplied by the sponsor.

2.4 Inclusion Criteria

2.4.1 Population

Adults aged over 18 years with diagnosed or suspected asthma or COPD who have never had spirometry performed or who have not had a spirometry assessment in the previous 18 months.

2.4.2 Interventions

The intervention is remote spirometry. Where necessary for Q3 we may considered other remote testing or assessment in people with respiratory conditions or home testing using digital technologies for other conditions.

2.4.3 Comparators

The comparator for Q1 and Q3 is spirometry conducted in a laboratory, clinic or practice setting. For Q2 the comparator is care as usual, no intervention or spirometry conducted in a clinic or practice setting.

2.4.4 Outcomes

For Q1 the relevant outcome is measurement accuracy. For Q2 the outcomes relate to participant health and include unscheduled use of health care including emergency department attendance and hospital admissions, in addition to symptoms (e.g. breathlessness, night waking, limitation of

activity) and quality of life. Medication and other management changes will also be considered. For Q3 the relevant outcomes are participant and personnel experience, and barriers and facilitators to use of remote spirometry.

2.4.4 Study Designs

In the first instance we include existing evidence syntheses. We also needed to consider primary research and we prioritised studies with the most robust methodological approaches for each question. These include randomised controlled trials including trials of “test-and-treat” approaches (Q2); well-conducted studies of measurement accuracy (Q1); and well-conducted qualitative or mixed-methods studies (Q3).

3. Results

3.1 Measurement properties (Question 1)

3.1.1 Evidence supplied by sponsor

The sponsor supplied a document reporting a literature review. This reported searching the Scopus platform using two terms and review of those abstracts obtained which were published within the previous four years. No other details of review methods were reported, meaning that we are unsure whether systematic review methods were employed following what appears to be a limited search.

We identified the published evidence cited in the review (with one exception [1]) two additional citations were for unpublished evidence. Evidence was not reported with any indication of study size or assessment of study limitations.

The sponsor's review considered the following:

i. Spirometry in primary care
ii. In-clinic validation of the remote spirometry versus two different systems MIR Spirolab (for asthma and COPD separately) and Jaegar Masterlab (for mixed lung diseases) iii) Spirometry at home versus laboratory spirometry (not supported)
iii. Spirometry at home versus laboratory spirometry (supported)
iv. Detection of treatment effects
v. Variations in spirometry measurements and their reliability across time and patient populations

We consider the evidence supplied to relate to Q1 only and focus on that relating to (ii) through (iv).

(ii) In-clinic validation

The submission presents validation data from two studies. These are looking at slightly different remote systems/devices (NuvoAir Next and NuvoAir-Smart). NuvoAir Next is a legacy system.

A cross-sectional study of NuvoAir Next (remote system) (de vs MIR spirolab (standard) was performed in a hospital outpatient clinic in Greece.(1) 150 people with lung disease (COPD, asthma or interstitial lung disease) and 50 healthy volunteers were enrolled. The study measured metrics including FEV_1 and FVC in absolute values and as % of predicted values and the ratio of FEV_1/FVC using both devices in clinic. The study enrolled patients relevant to this RES and to intended use of the device in practice and reported relevant outcomes. Although correlations were reported in the sponsor's submission they also used Bland Altman plots to look at the bias between the mean differences for the two devices. They reported "a small mean difference between the two devices with the majority of measurements being well within the limits of agreement". They did not look at the ability of the portable device to identify patients meeting particular diagnostic criteria on the

standard spirometry (sensitivity and specificity).

A cross-sectional study of the NuvoAir-Smart Spirometer vs the Jaegar Masterlab was performed in pulmonary function laboratories of two hospitals in Spain on patients with lung disease indications undergoing testing.(2) The study measured FEV[d] and FVC in absolute values and as % of predicted values and the ratio of FEV_1/FVC. The study found statistically significant differences between the values on the two spirometry methods; these were significantly lover for the two FVC measures and the absolute FEV_1 and significantly higher for the other measures. Bland Altman graphs showed homogenous distributions of measurements for FEV_1 and FVC but not for the ratio between them. The study also looked at the diagnosis of airway obstruction where 8/83 patients detected with convention spirometry would not have been detected with the portable spirometer using the FEV_1/FVC threshold of 0.7. The sensitivity of the remote device was 90.4% and the specificity 97.2% (based on 3/117 patients without airway obstruction identified as having it). It's not clear how relevant the patients in this study are to those seen in primary care for spirometry for asthma or COPD.

(iii) Unsupported home spirometry

The submission presents data from two published studies and some unpublished data.(3, 4) One study looked at almost 2,500 patients with COPD undergoing fluticasone withdrawal or remaining on fluticasone in an RCT.(4) Patients were trained and then undertook unsupervised daily home spirometry and periodic in clinic spirometry using the same device model (EasyOne). Mean home FEV_1 for the week in which clinic visits occurred was compared with the clinic measurement. FEV_1 was lower in-home measurements than in clinic, but consistently so across time, at a mean of 0.05L with good agreement on Bland Altman plots.

A second study shows post-hoc comparison of daily home spirometry with periodic in clinic spirometry in >400 participants aged ≥12 years with asthma in a multi-arm RCT of beclomethasone.(3) Devices used were not clearly reported. It was not clear that all participants contributed to this analysis. Adolescents aged 12-17 years were also considered as a separate subgroup. This study also found that mean FEV1 was lower in-home measurements than in clinic.

Data from these studies is directly relevant to people with asthma but may not be applicable to people with asthma not requiring inhaled steroids.

(iv) Supported home spirometry

The submission presents data from one study of 40 adult participants with a diagnosis of confirmed or suspected amyotrophic lateral sclerosis (ALS) or a closely related condition.(5) Spirometry using NuvoAir Next was compared with a clinic based device (MiniSpir) or another consumer-grade spirometer (SP10). The testing was conducted in clinic with the remote connection used to simulate

[d] We were unable to identify the intended reference given as follows: Stanzler and Stegenga, Resp. Therapy; 2019 15(1)

home testing. The study compared FVC measurements and ability to detect results below thresholds between 50% and 80% of expected results. FVC measurements were overestimated by remote spirometry and the overestimation was greater for higher % of predicted values. This was reduced by considering volumes rather than %. The ability to detect FVC below the 50% threshold was 100% for both sensitivity and specificity but these were lower – and for sensitivity much lower – for higher % thresholds.

3.1.1 Published evidence synthesis

A 2022 review looking at asthma management identified access to diagnostic spirometry as a barrier to care and suggested non–office-based digital spirometry as a potentially cost-effective method of addressing this.(6) Although using a systematic search this review lacked clear inclusion criteria and other indicators that it was truly systematic. The recommendation for digital spirometry is suggested despite the only study cited indicating that agreement between one spirometry measurement (trough FEV_1) taken in the clinic and at home is poor; this study recommended that home spirometry with the device used (the AM3 device) cannot be used as an alternative to clinic spirometry (using MasterScope device).(7) It was undertaken as a study within a phase III trial in 2434 adults with asthma which was uncontrolled despite combined inhaled steroid and LABA treatment. The lower and upper limits of agreement were -812 mL and 943 mL at baseline and -771 mL and 980 mL at week 24. Six percent of patients were outside these limits of agreement at both timepoints.

A 2018 systematic review (search date 2016) looked at remote respiratory assessments in people with COPD.(8) This review focused on regular assessments rather than a single diagnostic assessment or test but included two studies on novel assessments of respiratory function to identify people with COPD; neither of these used remote spirometry. One small feasibility study (14 participants) looked at the use of home spirometry analysis to classify people as having asthma or COPD.

An out-of-date review (2004) included four studies looking at the performance of three portable or semi-portable spirometers (Spirophone AG-SP, VM Plus, and Micro DiaryCard spirometer). (9) Summary data on measurement agreement with laboratory measures indicated variable results. The Spirophone AG-SP showed accuracy within ± 1% and validity ± 5 %; the VM Plus overestimated peak expiratory flow rate and showed reproducibility within ± 4% (not statistically significantly different from comparator); and the Micro DiaryCard showed variations between different devices of ±1.1 %; accuracy within ± 5% and tended to underestimate lung function. Studies were small (10 to 70 participants with COPD or asthma), and participants included healthy volunteers and children with asthma as well as adults with COPD or asthma.

All of these reviews would be considered to have high risk of bias under a ROBIS assessment.(10) This means that we cannot be confident that they have identified all the relevant evidence or that they have used methods designed to reduce error and bias in the review processes. None of the reviews assessed the methods used in the primary studies they included for reliability. This makes it difficult to determine the certainty of the evidence they represent. However all of the relevant included primary studies were small with the exception of the large CAPTAIN study, which found that the remote spirometry method assessed was not reliable on at least one measure. (7) With small studies it is more likely that additional research will change the conclusions from an evidence

synthesis; this is also the case where studies are not known to be well-conducted.

3.1.1 Primary research

Because we did not identify directly relevant published evidence syntheses which were both recent and reported use of rigorous methods we looked for relevant primary studies.

We found a highly relevant study (the COMPAIR study) of the NuvoAir Next spirometer which compared direct supervision of device use in clinic with virtual supervision at home. The study planned to enrol 68 adults aged ≥18 and ≤ 80 years with diagnosed asthma or COPD but without an exacerbation in the last 30 days or clinically significant/uncontrolled comorbidities considered likely to impact ability to participate. The study period is 8 days, during which two measurements will be taken at home and two in clinic for each participant. The values obtained from the two tests performed at home will be compared, allowing the calculation of test-retest variability. The study is a collaboration between the University of Nottingham and the sponsor of the innovation being assessed for this RES. The estimated study completion date is listed as May 2022 and the University of Nottingham website states that study data collection was completed in April 2022 but no results have been posted to clinicaltrials.gov.(11, 12) This study although not large will address the questions of replicability and comparability to practice-based testing of virtually supervised home spirometry using this device. The results of this study are likely to be informative for this RES.

We did not find any completed studies looking at the NuvoAir spirometry system which were not included in the sponsor's submission. We did identify a number of studies of other devices which were also indirectly relevant in other ways.

3.2 Effectiveness (Q2)

3.2.1 Evidence synthesis

Most of the research we identified looked at the use of regular remote monitoring, including spirometry, of people with diagnosed asthma or COPD for the purposes of disease management. This means it is not directly relevant to the question of the effectiveness of single-use remote spirometry for confirmation of diagnoses.

A 2022 review looking at asthma management identified access to diagnostic spirometry as a barrier to care, and suggested non–office-based digital spirometry as a potentially cost-effective method of addressing this.(6) Although using a systematic search this review lacked clear inclusion criteria and other indicators that it was truly systematic. The evidence cited for under-use of spirometry came from a 1999 survey of primary care doctors in Chicago; another study suggested that there is less use of spirometry among people in lower socioeconomic groups.

We identified two Cochrane reviews. One 2013 review looked at smartphone and tablet self-management apps for asthma,(13) and one 2021 review (search date 2020) looked at remote monitoring and consultations for people with COPD.(14) The 2021 review included 29 RCTs with over 5,500 people diagnosed with COPD.(14) Most of the interventions used monitoring over a period of

weeks and most used asynchronous review by healthcare professionals of remotely transferred data. Methods used to monitor participants included spirometry at home. The outcomes included mortality, exacerbations of disease, hospital admissions, quality of life and symptom measures. There was moderate certainty evidence that patients who have asynchronous remote monitoring plus usual care or as part of a multi-component intervention have a longer time before hospital readmission than those in the control group. All of the evidence for other outcomes and other comparisons was low or very low certainty. The 2013 review included only two studies both of which used an app for recording of data including peak flow measurements; neither used remote spirometry.(13)Other records identified were a protocol for which we could not identify a completed review (15) or were not published in English.(16)

The 2021 Cochrane review is well conducted and recent, and its assessments of the certainty of the evidence are very likely to be reliable. However, they are only indirectly relevant to people with asthma and to the use of spirometry as a single assessment for diagnostic purposes. We looked for relevant primary studies but did not identify any; impacts of remote spirometry on patient outcomes were not assessed in studies using it as a diagnostic rather than a continuous monitoring tool. In one small study in Australia, children with cystic fibrosis had remote spirometry instead of a clinic visit and this was reported to save patients a median travel time of >5.5 hours per visit; a finding which would be relevant to people living in remote rural areas of the UK and requiring spirometry for any reason.(17)

3.3 Patient experience (Q3)

We did not identify any relevant evidence syntheses for this question and therefore looked for relevant primary studies.

Because of the lack of directly relevant evidence syntheses we looked for primary studies. A study submitted by the sponsor looked at acceptability of remote supported spirometry of the NuvoAir device to both patients and their carers and therapists.(5) Scores out of five points were recorded for 15 items in general, FVC and maximal inspiratory pressure groups of questions. Mean scores were above 3 for all items; for patients/carers they were above 4. Qualitative responses were also reported. This was a simulated evaluation conducted in clinic and the patients were a group who may be very different from those who will be tested in primary care for asthma or COPD.

The study of the NuvoAir Next spirometer (the COMPAIR study, see Q1 above) used a standardised survey to evaluate participants' perceptions of home spirometry using this device with virtual supervision.(11, 12) The results of this study have not yet been reported but are likely to be very informative for Q1 and Q3 of this RES.

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The information in this report is correct at the time of printing.